SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Concept Mapping (Medium)
Assessment Tool Weightage: 35%

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QUESTIONS Total Marks: 50

Q.No	Question	Bloom's Level	Marks
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand	**5**
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand	**5**
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand	**5**
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand	**5**
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand	**5**
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand	**5**

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand	**5**
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand	**5**
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand	**5**
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember	**5**

Code of Conduct:

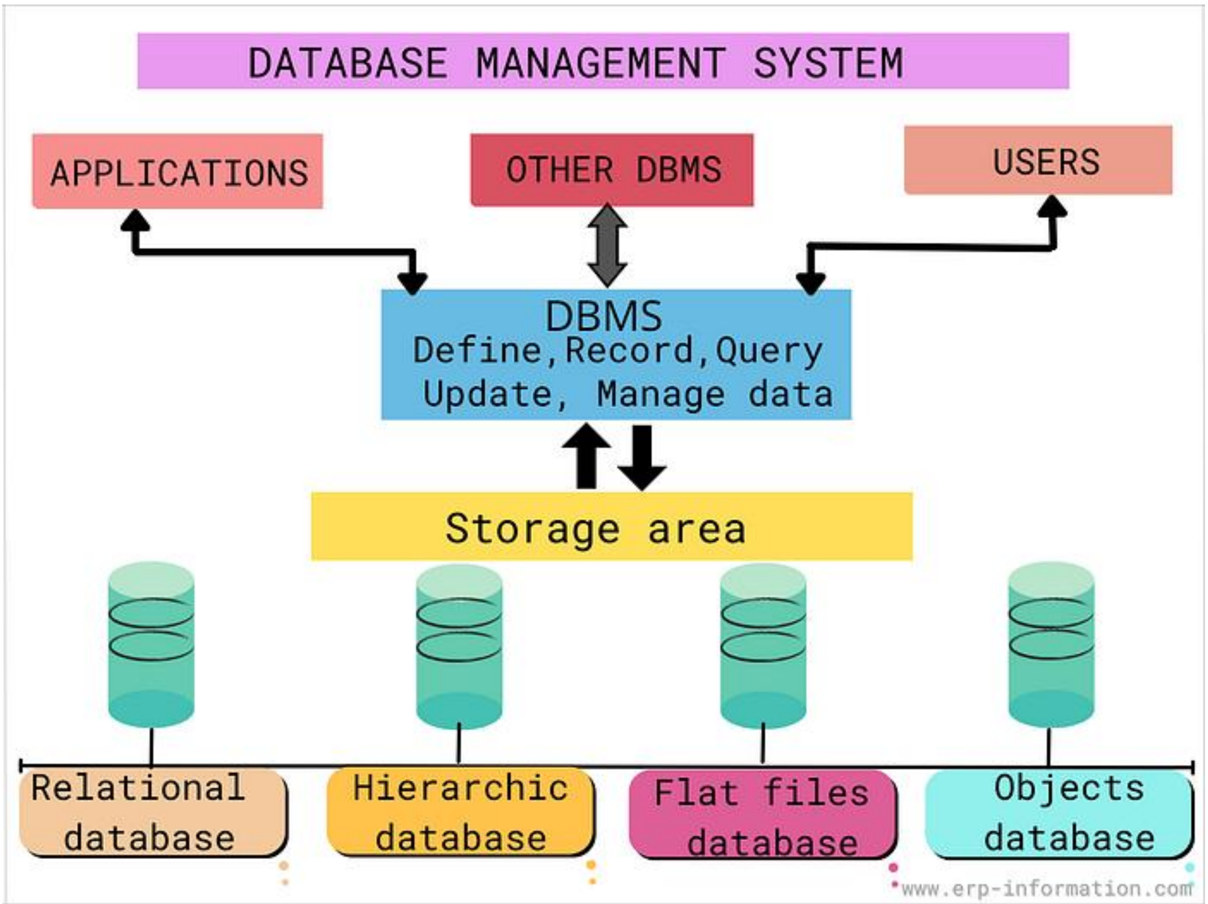
I KESHAVARAM L/192511253 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:KESHAVARAM L

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand



1. Concept Map: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship

Database Management System (DBMS)

- A software system used to create, store, retrieve, update, and manage databases.
- Uses a **Data Model** to organize and represent data.
- A **Data Model** defines how data is structured, stored, and related.
- An **Entity-Relationship (ER) Diagram** is used during database design.
- An **Entity** represents a real-world object such as Student, Employee, or Product.
- An **Attribute** describes the properties of an entity (e.g., Student_ID, Name, Age).
- A **Relationship** shows how two or more entities are connected (e.g., Student enrolls in Course).
- A **Schema** defines the overall logical structure of the database, including tables and relationships.
- An **Instance** represents the actual data stored in the database at a specific point in time.
- The DBMS uses the schema to manage database instances efficiently.

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2. Concept Map: ANSI/SPARC Three-Level Architecture

ANSI/SPARC Architecture

- Divides the database into three levels.
- **External Level** provides different views for different users.
- Each user sees only the required part of the database.
- **Conceptual Level** represents the complete logical structure of the database.
- It defines entities, relationships, and constraints.
- **Internal Level** describes the physical storage of data on disks.
- It includes file organization, indexing, and storage techniques.
- **Logical Data Independence** allows changes in the conceptual schema without affecting external views.
- **Physical Data Independence** allows changes in storage methods without affecting the conceptual schema.
- This architecture improves flexibility, security, and maintainability.

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3. Concept Map: File-Based System vs Database System

File-Based System

- Stores data in separate files.
- Causes data redundancy because the same data is stored multiple times.
- Leads to data inconsistency when files are updated separately.
- Makes data sharing difficult.
- Provides weak security.
- Backup and recovery are difficult.
- Data retrieval is slow and inefficient.

Database Management System

- Stores data in a centralized database.
- Reduces redundancy through normalization.
- Maintains consistency using constraints.
- Allows multiple users to access shared data.
- Provides security through authentication and authorization.
- Supports backup and recovery.
- Improves data integrity and performance.

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4. Concept Map: ER Diagram Components

ER Diagram

- Contains **Entities**.
- Includes **Weak Entities**, which depend on strong entities.
- Has different **Attribute Types**:
  - Simple Attribute
  - Composite Attribute
  - Multivalued Attribute
  - Derived Attribute
- Represents **Relationship Types**:
  - One-to-One
  - One-to-Many
  - Many-to-Many
- Uses **Participation Constraints**:
  - Total Participation
  - Partial Participation

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5. Concept Map: Hierarchical, Network, and Relational Data Models

Hierarchical Model

- Organizes data in a tree structure.
- Supports one-to-many relationships.
- Suitable for organizational hierarchies.

Network Model

- Organizes data as a graph.
- Supports many-to-many relationships.
- Suitable for interconnected data.

Relational Model

- Stores data in tables.
- Uses rows and columns.
- Supports SQL.
- Uses primary and foreign keys.

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6. Concept Map: Enhanced Entity-Relationship (EER) Model
EER Model

- Extends the basic ER model.
- Supports **Generalization**, which combines similar entities into a superclass.
- Supports **Specialization**, which divides a superclass into subclasses.
- Supports **Aggregation**, which treats relationships as higher-level entities.
- Helps model complex real-world applications.

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7. Concept Map: DBMS Software Architecture
User
→ Sends a query.
Query Processor
→ Parses and validates the query.
Query Optimizer
→ Chooses the most efficient execution plan.
Execution Engine
→ Executes the optimized query.
Storage Manager
→ Retrieves or updates data.
Database
→ Stores data permanently.
Result
→ Returned to the user.

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8. Concept Map: Data Models and Real-World Applications
Hierarchical Model

- Used in file systems and organizational structures.

Network Model

- Used in social networks and telecommunication systems.

Relational Model

- Used in banking systems, hospital management, libraries, colleges, and e-commerce applications.

Each model is suitable based on its structure and relationship capabilities.

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9. Concept Map: Schema, Instance, and Data Independence
Schema

- Defines the database structure.
- Changes rarely.

Instance

- Represents the current contents of the database.
- Changes frequently.

Logical Data Independence

- Allows changes in the conceptual schema without affecting user views.

Physical Data Independence

- Allows changes in physical storage without affecting the logical schema.

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10. Concept Map: Roles and Responsibilities in a DBMS Environment

Database Administrator (DBA)

- Creates and maintains the database.
- Controls security.
- Performs backup and recovery.
- Monitors performance.

Application Developer

- Designs database applications.
- Writes SQL queries.
- Develops user interfaces.
- Maintains software.

End User

- Accesses the database.
- Inserts, updates, deletes, and retrieves data.
- Uses applications to perform daily tasks.